BrightNest

Grounded after-hours help for parents

COMP6708 Mini Demo · **Group 15**

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seanfontaine.dev/COMP6708_A3_Demo

One sentence

1

A learning-centre assistant that

answers from centre truth,
acts with tools,
and hands off when humans must.

Training-free · RAG · hybrid routing · live on the public web

The problem

1

Imagine you run a tutoring centre.

- Parents message after hours: typhoons, fees, seats, trials, booking
- Staff cannot live on WhatsApp 24/7
- A wrong fee or hours answer costs trust — fast

So you hired Group 15

A couple of weeks later we built **BrightNest**: grounded answers, structured actions, honest hand-offs.

BrightNest

Not a chatbot that invents tuition from thin air.

Technique: training-free RAG + tools

1

Why training-free? Centre docs change weekly ⇒ fine-tuning is the wrong loop.

RAG (self-built)

1. Chunk markdown KB (`content/`)
2. Embed with DashScope
3. Store / query in Chroma
4. Lexical rerank → top-k
5. Generate with Qwen + citations

Retrieval score $s(q, c) = \mathrm{sim}(E(q), E(c))$ · inspector shows rank, s, full chunk.

Hybrid router

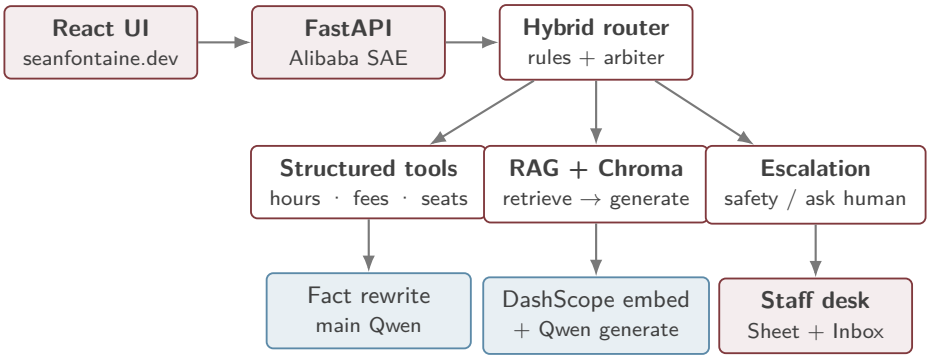
- Hard rules: typhoon, fees, seats
- Soft prototypes for paraphrase
- Cheap LLM review *only if* low confidence
- Else: `high_confidence_skip`

tools for truth **RAG for narrative**

escalate when human

Architecture

1



Inspector shows: which rule fired · arbiter skip/call · RAG ranks & scores · session memory

What you will see live

2

Parent chat

- Messaging surface + probes
- Bookable seat cards

System inspector

- Router decision (rules fired / idle)
- RAG chunks: expand, rank, score
- Session memory: expand turns

Staff desk

- Follow-up sheet (soft gaps)
- Gmail-style urgent inbox

5-minute rule

Slides cue. We **operate** the system.

Open the live demo fullscreen. Reset once.

Demo beats (cue card)

2

Beat	Show	Point at
1	Typhoon	Router: rule 1.0 · arbiter skip · RAG skip
2a	Phonics fee	Structured fees tool (not RAG)
2b	RAG · Private 1:1	Expand chunk · rank · score · top-k
3	Saturday seats → Book	Live register + cards
4	Scuba (unknown)	Follow-up **sheet** row
5	Safety allergy call	Staff **inbox** urgent mail

Beats 1→5 without reading bullets. Narrate the inspector.

Impact & why it works

3

Impact

- 24/7 triage without inventing policy
- Tools for hours / fees / seats
- RAG for programme narrative
- Humans when it matters

Creativity

- Hybrid router + confidence-gated review
- Staff desk = Sheets + Gmail clones

Correctness cues

Visible rule map + decision path

Citations with retrieval scores

Urgent safety inbox + email draft

Off-topic refuse (no fake todo)

Evaluation

Live path correctness · visible RAG · latency feel.
No formal user study required.

Thank you

3 Impact

Grounded answers. Structured actions. Honest hand-offs.

RAG · tools · staff desk

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Q&A

Thank you for listening!
Your feedback will be highly appreciated!